HONGIK UNIVERSITY

2026년 2월 26일

EoMT ANE(Apple Neural Engine)

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목표

목표 (~2월)

iOS 환경에서 EoMT 모델 panoptic segmentation 추론 및 최적화

Panoptic segmentation: 어떤 사물인지, 개별 객체인지를 동시에 구분

- Coreml ANE(Apple Neural Engine) 활성화

Related Work:

MobileSAM, EdgeSAM: 경량화된 Encoder를 활용한 on-device 모델 사례

진행사항

- 사용 기기: iphone 13 (A15) → iphone 17 pro (A19 Pro)

- 기존의 문제점: FP32로 연산하도록 EoMT 모델을 저장했을 때 ANE(Apple Neural Engine)의 미사용. (CPU와 GPU만 사용)
  - FP32로 저장된 모델 - ANE 미사용
  - FP16로 저장된 모델 - ANE 사용

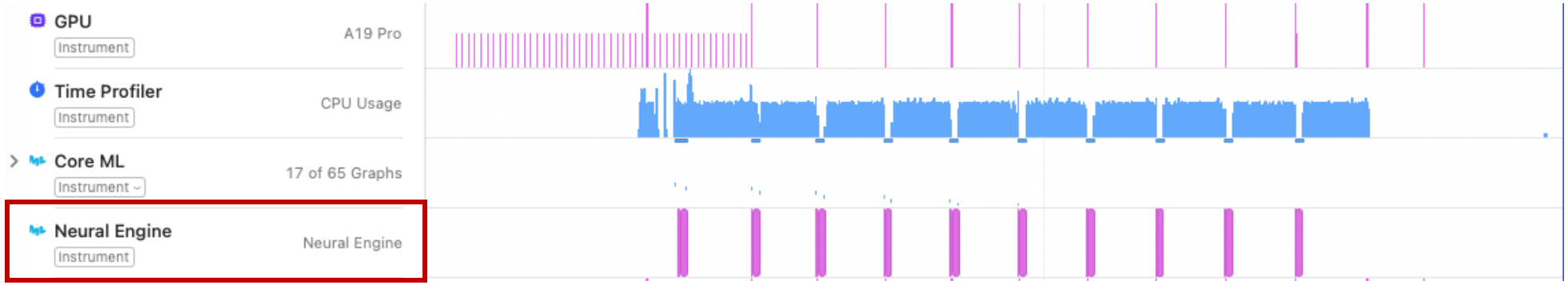
To do

- FP16으로 저장된 모델에서 segmentation 정확도 판단

진행사항

FP32

FP16



진행사항

FP32 EoMT model

Precision

Compute
Mixed (Float32, Int32)

Storage
Float32

compute_precision=ct.precision.FLOAT32

FP16 EoMT model

Precision

Compute
Mixed (Float16, Float32, Int32)

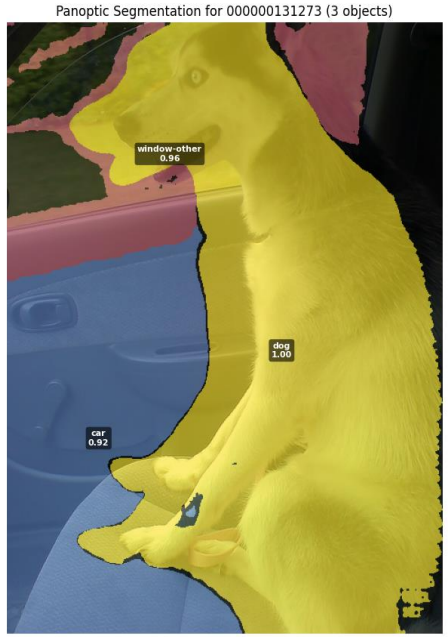
Storage
Float16

compute_precision=ct.precision.FLOAT16

진행사항

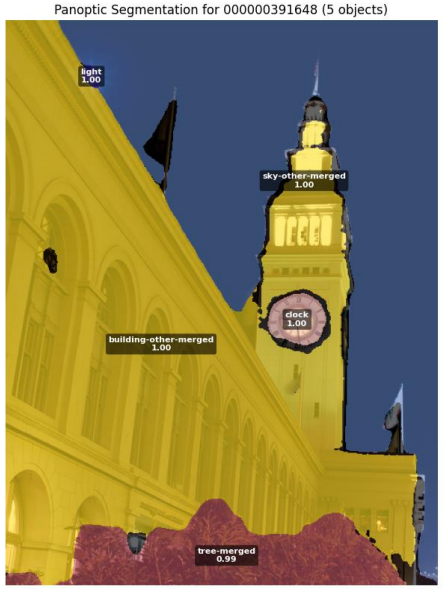
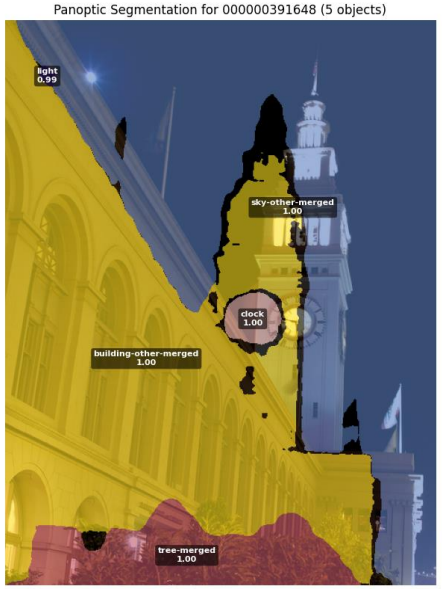
FP32 segmentation

FP16 segmentation



진행사항

FP32 segmentation



FP16 segmentation